

Lecture Unit 1.1

What is HPC and what
is it used for?

What is HPC?

“ *High Performance Computing most generally refers to the practice of aggregating computing power in a way that delivers much higher performance than one could get out of a typical desktop computer or workstation in order to solve large problems in science, engineering, or business.* ”

<http://insidehpc.com/hpc-basic-training/what-is-hpc/>

What is HPC?

“When someone says HPC it means something really specific to traditional HPC folks; it’s tightly coupled, we’ve got some sort of low latency interconnect, parallel file systems, designed to run high performance, highly scalable custom applications. But today, this has changed. HPC has come to mean pretty much any form of scientific computing and as a result, its breadth has grown in terms of what kind of applications we need to support.” – Gregory Kurtzer, Singularity (HPC container software).

<https://www.hpcwire.com/2017/12/21/five-surefire-maybe-hpc-predictions-2018/>

“Traditional” Applications of HPC

- Traditionally used in physical sciences, engineering and applied mathematics
- Realistic problems frequently generate systems of equations which cannot be solved analytically (e.g. Navier-Stokes equations in fluid dynamics)
- Often we can calculate approximate numerical solutions which answer the question we are addressing
- This can be a computationally expensive process

The Met Office

The Met Office supercomputing system is one of the most powerful in the world dedicated to weather and climate.

This hugely complex Government IT project was successfully completed in December 2016, delivered early and on budget. Some fifteen times larger than its predecessor, this major science infrastructure investment has enabled the UK to continue to lead the world in weather, climate and environmental science high-performance computing.

Our supercomputer consists of three main systems – an identical pair of machines and a single larger system in a purpose-built data centre nearby. The twin identical machines provide a highly resilient capability for running time-critical operational weather forecasts, whereas the third system provides research, development and collaboration capabilities.

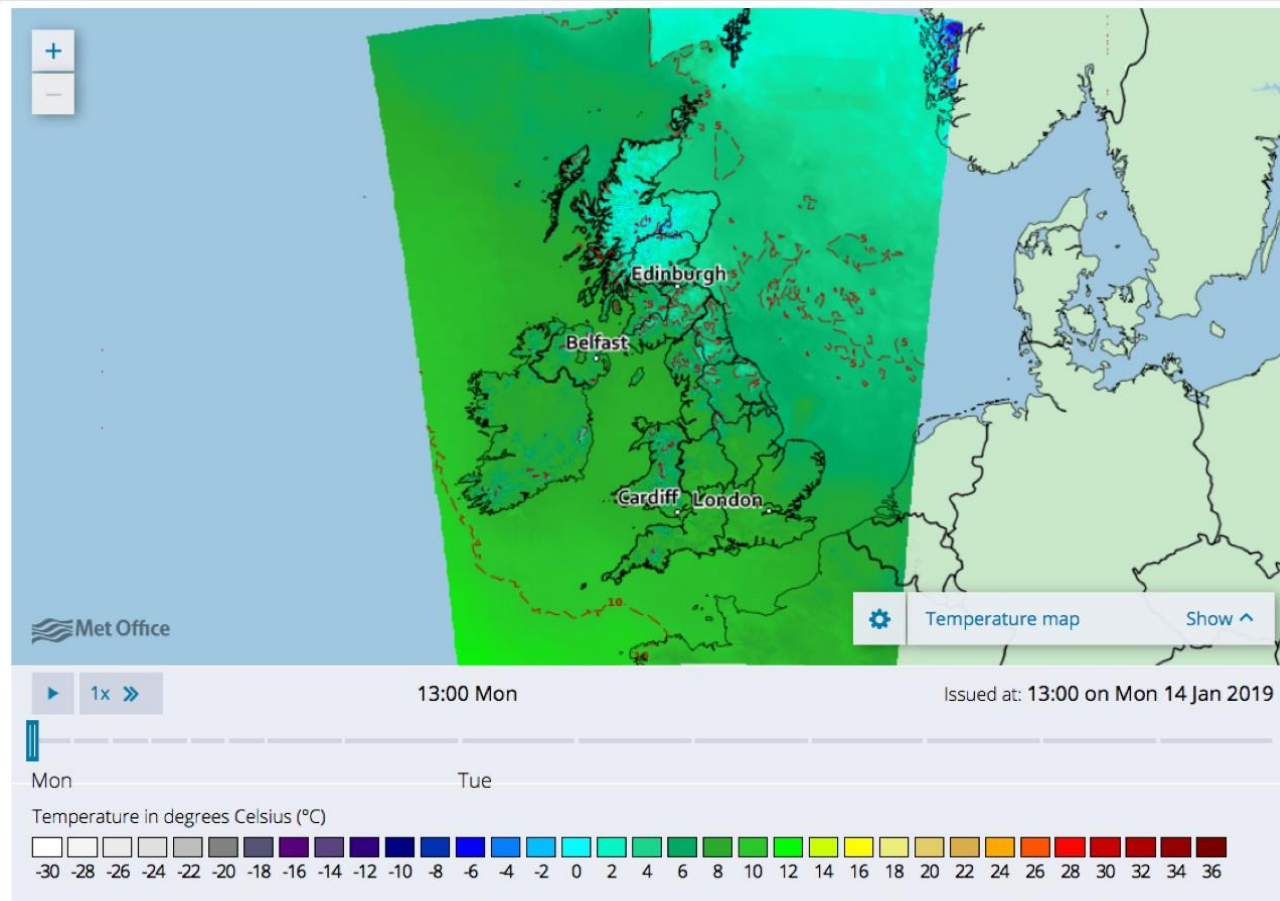
At the time of installation all three phases of our current supercomputing system appeared in the top 50 of the world's most powerful computers. The Met Office's supercomputer capability currently remains one of the most powerful in the world dedicated to weather and climate.



- The Met Office are major users of HPC in the UK
- HPC is vital for:
 - making weather forecasts
 - studying climate change
 - oceanography

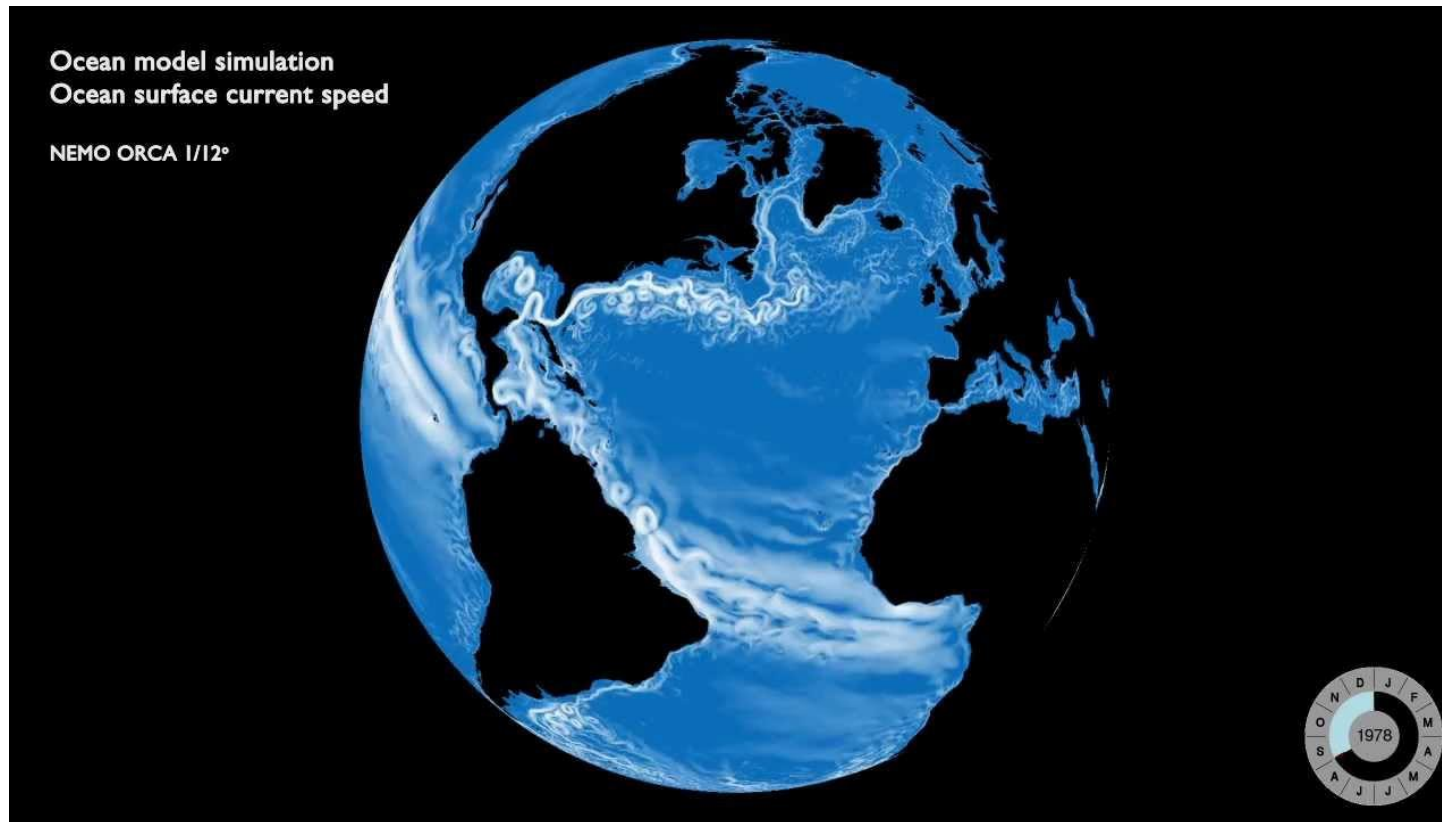
<https://www.metoffice.gov.uk/about-us/what/technology/supercomputer>

Weather Forecast



<https://www.metoffice.gov.uk/public/weather/forecast/map>

Oceanography



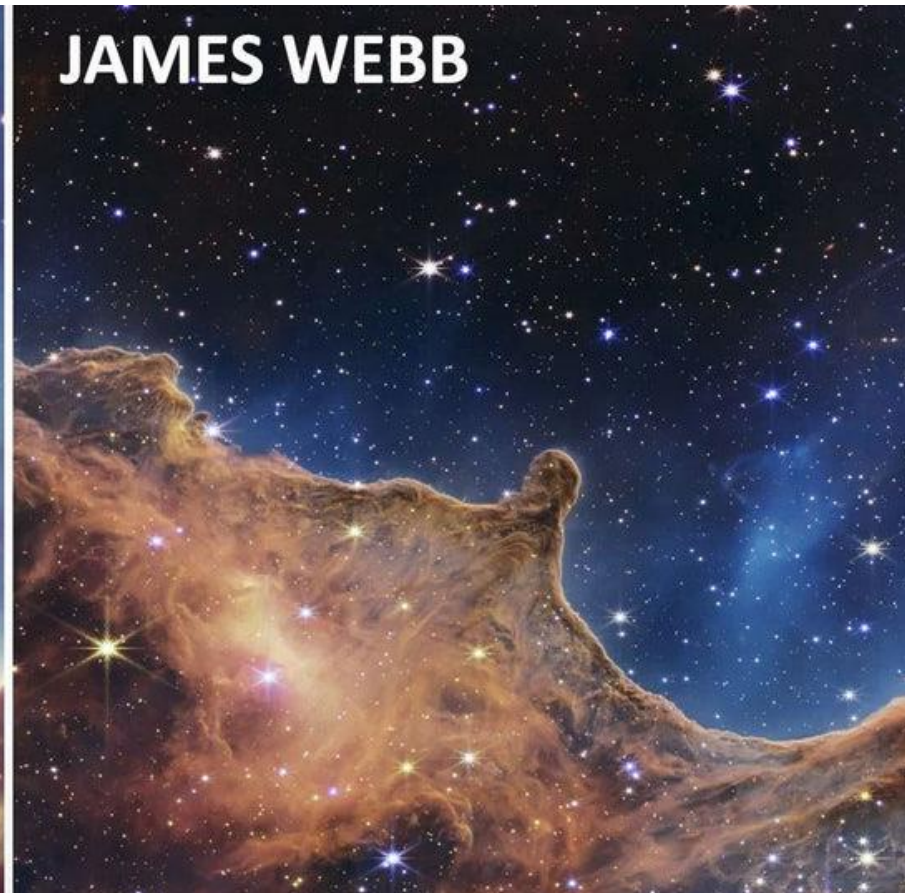
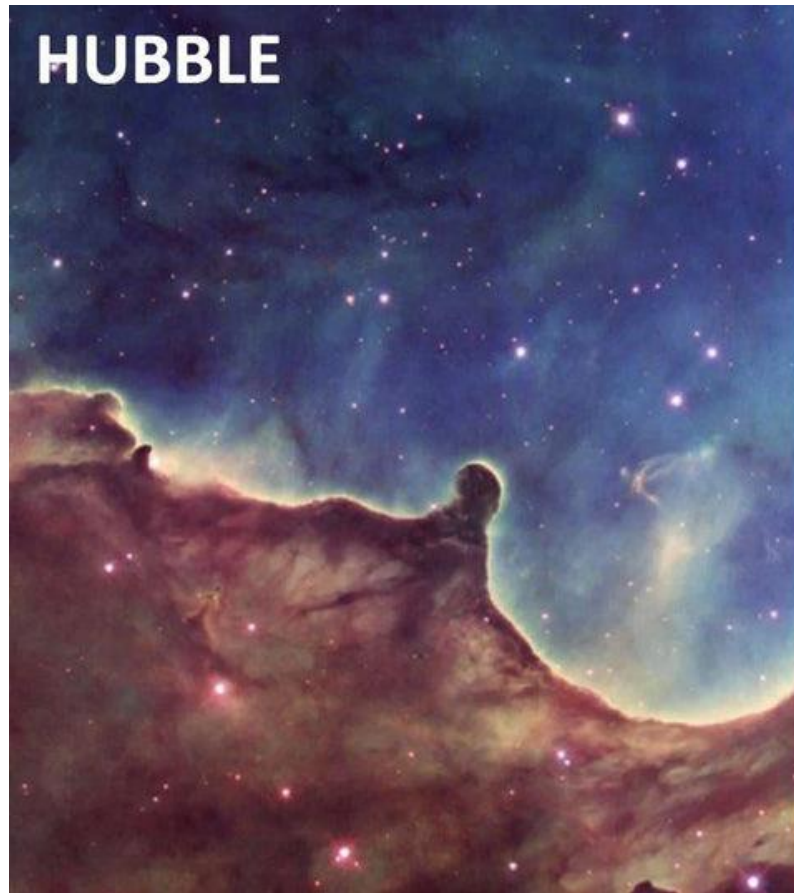
<https://noc.ac.uk/news/nemo-movie-goes-live>

Astrophysics: Star Formation



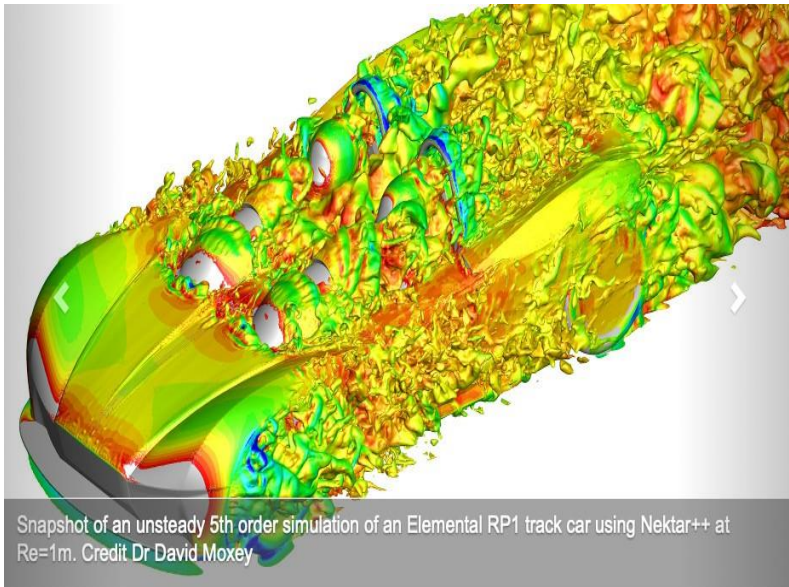
<http://www.astro.ex.ac.uk/people/mbate/Cluster/cluster3d.html>

Astrophysics: James Webb Vs Hubble



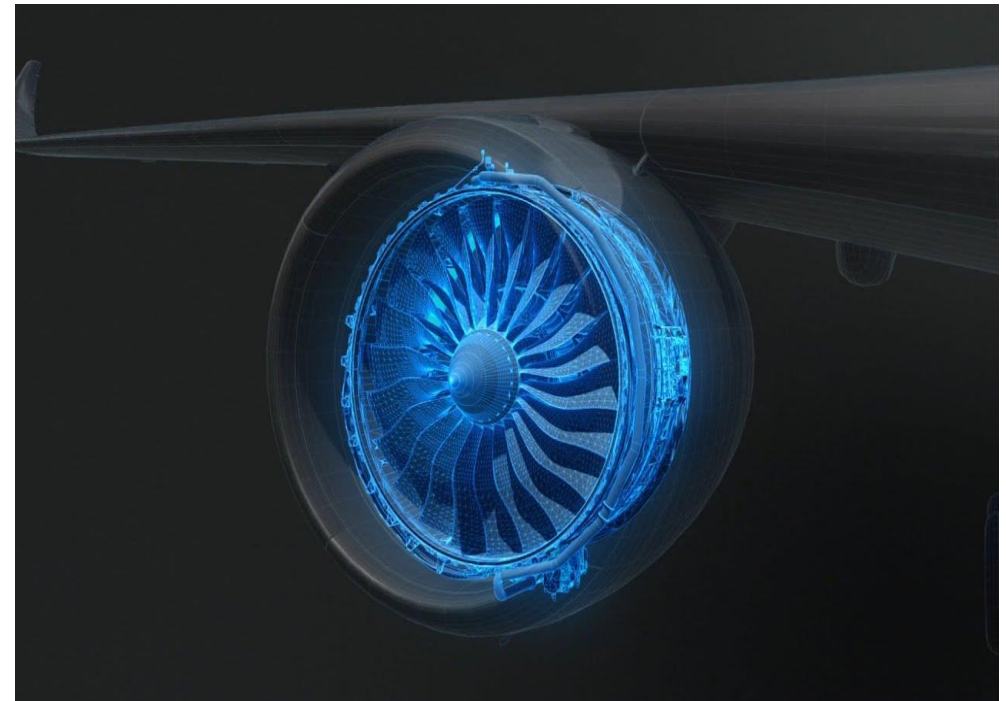
<https://astrophotographylens.com/blogs/astro/james-webb-telescope-vs-hubble>

Computational Engineering and Digital Twins



Computational Engineering is the use of computers to simulate and understand engineering problems. In the Computational Engineering group we develop and use a range of techniques - Computational Fluid Dynamics, Finite Element Analysis, Surrogate Modelling, Optimisation and Machine Learning - and apply these to real world problems in industry and medicine.

Our work is supported by a range of research grants and we often work with industrial partners - take a look at some of the collaborations and projects currently active.



<http://emps.exeter.ac.uk/engineering/research/ourresearch/computational-engineering/>

Other Physical Sciences Applications

- Have looked briefly at three applications which use Computational Fluid Dynamics (CFD)
 - Weather, climate, and oceanography (geophysics)
 - Astrophysics
 - Engineering
- There are other HPC applications in the physical sciences and engineering (e.g. electronic structure, molecular dynamics)

Life Sciences: Genomics

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The Exeter Sequencing Service at the University of Exeter is a core facility providing state-of-the-art genomics and bioinformatics analysis. We specialise in providing a bespoke service to researchers in academia and industry.

Life Sciences: Genomics

- Life sciences now have extensive computing requirements (e.g. genomics)
- More data orientated than traditional HPC
- Reflected in use of our local HPC system (Isca)

Isca research papers

Journal articles

Cunliffe AM, McIntire CD, Boschetti F, Sauer KJ, Litvak M, Anderson K, Brazier RE (2020). Allometric Relationships for Predicting Aboveground Biomass and Sapwood Area of Oneseed Juniper (*Juniperus monosperma*) Trees. *Frontiers in Plant Science*, 11. [Abstract](#). [Author URL](#). [Full text](#). DOI.

De Franco E, Caswell R, Johnson MB, Wakeling MN, Zung A, Dũng VC, Bích Ngọc CT, Goonetilleke R, Vivanco Jury M, El-Khateeb M, et al (2020). De Novo Mutations in EIF2B1 Affecting eIF2 Signaling Cause Neonatal/Early-Onset Diabetes and Transient Hepatic Dysfunction. *Diabetes*, 69(3), 477-483. [Abstract](#). [Author URL](#). [Full text](#). DOI.

Pilling LC, Jones LC, Masoli JAH, Delgado J, Atkins JL, Bowden J, Fortinsky RH, Kuchel GA, Melzer D (2020). Low Vitamin D Levels and Risk of Incident Delirium in 351,000 Older UK Biobank Participants. *Journal of the American Geriatrics Society* [Full text](#). DOI.

Jones G, Pilling LC, Kuo C-L, Kuchel G, Ferrucci L, Melzer D (2020). Sarcopenia and Variation in the Human Leukocyte Antigen Complex. *J Gerontol a Biol Sci Med Sci*, 75(2), 301-308. [Abstract](#). [Author URL](#). [Full text](#). DOI.

Kuo C-L, Joaquim M, Kuchel GA, Ferrucci L, Harries LW, Pilling LC, Melzer D (2020). The Longevity-Associated SH2B3 (LNK) Genetic Variant: Selected Aging Phenotypes in 379,758 Subjects. *J Gerontol a Biol Sci Med Sci*, 75(9), 1656-1662. [Abstract](#). [Author URL](#). [Full text](#). DOI.

Ruth KS, Day FR, Tyrrell J, Thompson DJ, Wood AR, Mahajan A, Beaumont RN, Wittemans L, Martin S, Busch AS, et al (2020). Using human genetics to understand the disease impacts of testosterone in men and women. *Nature Medicine*, 26(2), 252-258. [Full text](#). DOI.

Casanova F, Tyrrell J, Beaumont RN, Ji Y, Jones SE, Hattersley AT, Weedon MN, Murray A, Shore AC, Frayling TM, et al (2019). A genome-wide association study implicates multiple mechanisms influencing raised urinary albumin-creatinine ratio. *Hum Mol Genet*, 28(24), 4197-4207. [Abstract](#). [Author URL](#). [Full text](#). DOI.

Wright CF, West B, Tuke M, Jones SE, Patel K, Laver TW, Beaumont RN, Tyrrell J, Wood AR, Frayling TM, et al (2019). Assessing the Pathogenicity, Penetrance, and Expressivity of Putative Disease-Causing Variants in a Population Setting. *American Journal of Human Genetics*, 104(2), 275-286. [Abstract](#). [Full text](#). DOI.

<https://www.exeter.ac.uk/researchcomputing/researchresults/iscaresearchpapers/>

Unit Summary

- HPC is widely used in science and engineering
- In the past physical sciences and engineering were the major HPC users
- Life sciences now are now major user of HPC for example to analyse genomics data
- See [SAB Chapter 1.2](#) and <https://epsrc.ukri.org/newsevents/pubs/impactofnationalhpc/> for more examples of applications and their impact